Xiaoqing PAN

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Research Center Borstel - Leibniz Lung Center (FZB), Borstel

Re: Postdoctoral Researcher

Dear Prof. Barber,

My name is Xiaoqing, I just finished my PhD study at Hans-Knöll-Institute in Jena, Germany. I expect to have my defense in July 2026. I am writing to apply for the open postdoc position in your group.

I studied microbiology in the context of human fungal pathogens. My PhD project was focusing on double-stranded RNA metabolism in both humans and pathogenic fungi. I was elucidating how *Aspergillus fumigatus* produces small RNAs, how dicer-like proteins process dsRNA substrates, and how excess dsRNA arrest fungal growth in opposite of your work of dsRNA virus promoting fungal fitness. I also combined bioinformatics (sRNA-seq and mRNA-seq analyses) and experimental validations (qPCR as well as Northern blot) to characterize the small RNA landscape and the contribution of RNAi machinery to fungal biology. This work has resulted in two publications on *RNA* and *Molecular microbiology.* In parallel, I was involved in another project investigating host transcriptomic responses. Through characterizing transcriptomic responses after fungal stimulations with a focus on host alternative splicing and A-to-I RNA editing, we surprisingly found out that *Rhizopus oryzae can* induce A-to-I editing as well as disrupt host RNA splicing. I contributed to analyse multiple RNA-seq cohorts, DEG analysis via DESeq2, alternative splicing analysis via rMATS-Turbo, RNA structure analysis via SHAPE-Map and RNA editing analysis. Through my PhD training, I ‘ve been equipped with solid wet-lab skills, such as fungi infecting human cell assay, microbial culturing, DNA/RNA extraction, and gained experiences on bioinformatics.

I would like to develop an academic career in fungal research. I believe that postdoctoral training is essential for my career development. I find the prerequisites of the postdoc position in your group regarding “pH and salinity affecting fungal pathogenicity” align well with my background. For instance, I worked with pH-responsive transcription factor pacC in *A. fumigatus*, under the hypothesis that its mRNA is targeted by an sRNA. I wish in the future I can Investigate the evolutionary bridge between environmental pH adaptation and the ability of these pathogens to thrive in the acidic tissues of diabetic patients. My expertise lies in the host-fungal system at the molecular level, which is essential for understanding human fungal infections. I have knowledge of opportunistic fungal pathogens, including environment fungus *A. fumigatus* and commensal fungus *C. albicans.* I have experience in analyzing transcriptomic data in an HPC environment, and I am interested in expanding my skillset to other omics. The opportunity to dive into multi-omics data analysis in this project will help me to develop more computational skills.

My skillset on DNA technologies will complement together with your expertise. This is realistic since genomic analysis also relies on high-throughput sequencing technology. Besides, I began to investigate host response in my PhD, which created my interest in better understanding human interferon response. While you will work closely with the University Hospital Schleswig-Holstein, I believe my knowledge of immunology might be helpful in the future. Moreover, I am seeking an international collaborative work environment, as I find integration into a vibrant team important for my development and performing high-quality science. I noticed your group includes members with diverse backgrounds, and I believe that bringing my perspective from fungal genetics and host-pathogen immunology could contribute fresh insights to related projects in the group.

Thank you for your time and consideration.

Best,

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SUMMARY

RNA biologist and computational transcriptomics researcher with expertise in high-throughput sequencing, gene regulation, and RNA-mediated cellular responses. Experienced in integrating molecular biology with bioinformatics to study small RNAs, RNA interference, alternative splicing, and RNA editing across host and microbial systems. Proficient in RNA-seq analysis, reproducible computational workflows (R, Bash), and sequencing-based approaches for investigating post-transcriptional regulation.

TECHNICAL SKILLS

- **Molecular Biology**: sRNA cloning, NGS library preparation, GUS reporter assays, luciferase and GFP reporter assays, Gibson Assembly Cloning, Genetic Engineering, Northern Blot, qPCR, PCR, immunostaining, LC-MS, cultivation of microorganisms (BSL-1/2), plant cultivation and tissue preparation
- **Bioinformatics**: RNA-seq, sRNA-seq, differential expression, alternative splicing, small RNA characterization and target prediction, data visualization, HPC (Slurm), UCSC, SwissProt, KEGG, NCBI
- **Programming**: R, Python, Bash, Git, Docker, Linux
- **Language**: Chinese (native), English (fluent), German (B1)

EXPERIENCES

RNA landscape of human fungal pathogen *Aspergillus fumigatus*

- Characterized RNAi and tRNA fragments processing pathways in *A. fumigatus* across developmental stages (conidia, mycelium), revealing morphotype-specific RNA regulatory mechanisms
- Analyzed differential expressed genes on mRNA-, sRNA-seq datasets, uncovering post-transcriptional regulatory networks controlling fungal development and stress response.
- Validated genes of interest (sRNA, tRNA fragments, sRNA targets) via qPCR, northern blot

Human PBMC transcriptome changes upon fungal stimulations

- Analyzed pathogen-specific transcriptomic signatures in human immune cells responding to fungal pathogens, characterized differential expressed genes, alternative splicing, A-to-I RNA editing, pathogen recognition receptor pathway activation
- Built HPC pipeline, wrote custom R scripts for multi-condition cohort analysis

Human neutrophils exporting extracellular RNA upon *A. fumigatus* confrontation

- Analyzed EV-associated sRNA, tRNA, and mRNA cargo from primary human neutrophils during fungal challenge

Non-coding RNA analysis of rice-*Xanthomonas oryzae* interaction

- Studied how T3SS-dependent bacterial effectors from *Xanthomonas oryzae* shape host non-coding RNA regulation; performed GUS reporter assays, plant cultivation, tissue preparation, LC-MS, qPCR, and PCR
- Engineered overexpressed and knockout mutants of circular RNA and microRNA in rice, established genetic tools that enabled investigation of ncRNA-mediated regulation

Teaching

- Microbiology and Molecular Biology, MMB005, 2022-2025. Supervised master students investigating gene regulation together with Dr. Blango and other colleagues.

EDUCATION

PhD | Microbiology, Friedrich Schiller University, Germany (supervisor: Dr. Matthew Blango) 2026
Thesis: RNA regulation in host-fungal interactions: investigation of the host epitranscriptome and fungal small RNAs

MSc | Bioengineering, Chinese Academy of Sciences, China (supervisor: Dr. Kuaifei Xia) 2021
Thesis: Non-coding RNAs ciR133 and miR168a-5p play roles in response to Xoo and abiotic stress in rice, respectively

PUBLICATIONS

- **Pan X***, Kelani AA*, Schrettenbrunner L, Prabakar S, Israni B, Blango MG. (2026). An improved description of the small RNA landscape of the human fungal pathogen *Aspergillus fumigatus*. Mol Microbiol 1-12.
- **Pan X***, Bruch A*, Blango MG. (2024). The past, present, and future of RNA modifications in infectious disease research. ACS Infect Dis 10(12), 4017-4029.
- Xia K*, **Pan X***, Chen H, Xu X, Zhang M. (2023). *X00*-responsive transcriptome reveals the role of the circular RNA133 in disease resistance by regulating expression of *OsARAB* in rice. Phytopathol Res 5, 22.
- Xia K*, **Pan X***, Zeng X, Zhang M. (2023). Rice miR168a-5p regulates seed length, nitrogen allocation and salt tolerance by targeting *OsOFP3, OsNPF2.4*, and *OsAGO1a*, respectively. J. Plant Physiol 208, 153905.
- Kelani AA, Bruch A, Rivieccio F, Visser C, Krüger T, Weaver D, **Pan X,** Schaeuble S, Panagiotou G, Kniemeyer O, Bromley MJ, Bowyer P, Barber AE, Brakhage AA, Blango MG. (2023). Disruption of the *A. fumigatus* RNA interference machinery alters the conidial transcriptome. RNA 29(7), 1033-1050.

CONFERENCES

- **2025 - Oral presentation:** German speaking Mycological Society (DMykG), Germany
- **2025 - Poster:** Gordon Research Seminar - RNA Editing, Italy
- **2024 - Oral presentation:** Jena RNA Club Symposium, Germany
- **2024 - Poster:** RNA Society Meeting, UK
- **2023 - Poster:** EMBL - The Non-coding Genome Symposium, Germany

AWARDS & SCHOLARSHIP

- DAAD Scholarship 2021 - 2025
- Jena School for Microbial Communication (JSMC) travel grant 2023

REFEREE

- Dr. Matthew Blango (matthew.blango@leibniz-hki.de)
- Dr. Olaf Kniemeyer (olaf.kniemeyer@leibniz-hki.de)